Akshay Paghadar

Electronics Engineer | Embedded Systems & Robotics | IoT Design

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Professional Summary

Electronics Engineering graduate (B.Tech, 2026) specializing in embedded firmware, PCB design, and IoT product development — taking hardware from schematic to field deployment. National award winner (**Robofest 4.0**) and holder of **4 filed patents (2 granted)** for an autonomous underwater vehicle that placed **6th at SAUVC 2025, Singapore**, against 20+ international teams. Proven ability to deliver reliable, low-cost embedded solutions for agriculture, industry, and marine robotics. Seeking embedded systems, firmware, or robotics engineering roles.

Education

Gujarat Technological University — 2022 – 2026

B.Tech in Electronics Engineering — CGPA: 7.68 / 10.0 — *Vallabh Vidyanagar, India*

Technical Skills

Programming: C, C++, Python (Basic)

Embedded & Hardware: Embedded Systems Design, Microcontroller-Based Systems, PCB Design, Sensor Integration

Wireless & Communication: LoRa Wireless Communication, GSM, IoT Connectivity Protocols

Robotics & Navigation: Autonomous Navigation, LiDAR-Based Obstacle Avoidance, 6-DOF Motion Control

VLSI / Digital Design: RTL-to-GDSII Flow, Digital SoC Design & Planning (Open-Source EDA Tools)

Professional: Project Management, Technical Leadership, Cross-Functional Teamwork, Technical Communication

Experience

Instruments Universal — 2026

Industrial Intern — *Vadodara, India*

- Completed a 5-month industrial internship gaining applied exposure to instrumentation and industrial electronics workflows.

Monk9 Tech Private Limited — 2024 – 2025

VLSI Design Intern — *Rajkot, India*

- Executed digital SoC design and planning using open-source EDA tools, covering floorplanning and design-stage verification.
- Implemented a complete RTL-to-GDSII physical design flow using open-source tools, strengthening foundations in digital IC design.

Key Projects

Autonomous Underwater Vehicle (AUV) | *Team Lead — Robofest 4.0 & SAUVC 2025* — *2024 – 2025*

- Designed and built a cost-effective underwater robot with LoRa-based wireless communication, real-time surveillance, and 6-DOF maneuverability; secured a **design patent** and a **utility patent** in India for its innovations.
- Achieved **6th place** among international teams at SAUVC 2025, Singapore, and **Runner-Up** at the Robofest 4.0 National Competition.

Autonomous Kiwi Bot | *Independent Project* — *2025*

- Developed an autonomous mobile robot using LiDAR-based navigation and real-time obstacle avoidance algorithms; co-authored a technical paper documenting the methodology.

Smart Three-Phase Motor Automation System | *Final-Semester Industrial Defined Project* — *2025 – 2026*

- Engineered a low-cost automation system for three-phase motors with IoT-based real-time monitoring, fault detection, and remote control for agricultural and industrial use.
- **Won the Project Expo** for demonstrated practical impact and system reliability.

Additional Projects: GPS Quadcopter Drone, Smart RC Car, Digital Clock without Microcontroller, Smart IoT-Based Dustbin, Fully Automatic Water Level Controller, Contactless Phase Wire Detector, Smart Door Lock System — *more at akshaypaghadar.vercel.app/projects*

Publications & Patents

- **Paper:** Low-Cost Smart Three-Phase Motor Automation and Real-Time Monitoring Solution for Agriculture and Industry (ICAIIE-2026).
- **Paper:** Autonomous KIWI Bot with LiDAR-Based Navigation and Obstacle Avoidance — IJSRST, Vol. 12, Issue 2 (2025). DOI
- **Design Patent (447676-001):** Granted in India for an underwater robot with novel ornamental and functional features.
- **Utility Patent (202521018561 A):** Published in India for an underwater robot application enhancing marine exploration, inspection, and data collection.
- **Design Patent (494109-001):** In Process — application for a second-generation underwater robot design.
- **Utility Patent (202621027302):** In Process — application for extended underwater robot functionality.

Achievements, Leadership & Languages

- Awarded the **Gaurav Puraskar** and an **INR 10 Lakh** (approx. USD 12,000) cash prize for innovation in underwater robotics at Robofest 4.0 (2024–2025).
- Represented **India** at SAUVC 2025, Singapore — a global autonomous underwater vehicle competition.
- **National Cadet Corps (NCC):** Earned the Cadet rank and passed the "B" Certificate examination (2022–2024).
- **Languages:** Gujarati (Native) | Hindi (Fluent) | English (Professional Working Proficiency)